

2026 IEEE SUSTAINABLE POWER AND ENERGY CONFERENCE

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SPECIAL SESSION 5 (SS5) • EIPCES

Emerging Intelligence and Power Conversion for Electrified Systems

— Call for Papers —

Organized and Co-Chaired by:

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Session Overview:

This session focuses on the convergence of intelligent computing and advanced power conversion technologies for next-generation electrified energy systems, addressing renewable energy integration, transportation electrification, energy storage, and grid-interactive applications. It encourages contributions on AI/ML-enabled converter design and control, wide-bandgap devices, bidirectional high-gain converters, EV charging, V2G/V2H, and intelligent energy management — bridging power electronics with emerging applications in electric mobility and flexible energy infrastructure.

Topics of Interest Include, but Are Not Limited to:

- AI/ML and data-driven techniques for power electronic systems
- Advanced DC-DC, DC-AC, AC-DC, and AC-AC converter topologies
- High-gain, high-efficiency, high-power-density bidirectional converters
- Intelligent control, optimization, and predictive control of converters
- Wide-bandgap (SiC/GaN) devices and advanced switching techniques
- EV charging and V2G/V2H bidirectional systems
- Battery energy storage systems and battery-converter interfaces
- Microgrids, distributed energy resources, and grid-interactive converters
- Digital twins, condition monitoring, and predictive maintenance
- Reliability, thermal management, and lifecycle considerations

Important Dates:

Paper Submission Opens	25 August 2026
Paper Submission Deadline	30 September 2026
Acceptance Notification	15 October 2026
Registration Due	1 November 2026
Conference Dates	4–6 December 2026

Format: Half-day technical special session, targeting ~8–10 peer-reviewed papers. Author guidelines and peer-review standards will follow the regular iSPEC 2026 paper submission process. Accepted and presented papers will be submitted for inclusion in IEEE Xplore, subject to IEEE Xplore's scope and quality requirements.

Submission Portal: <https://ispec2026.org/submission>

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